

Overview:

Intuilize Inc. is a leading firm specializing in machine learning (ML) and change management solutions aimed at optimizing pricing and inventory strategies to enhance profitability for wholesalers. Many midsize wholesalers across the United States and Canada encounter challenges in effectively managing their pricing strategies due to limited resources in personnel, expertise, and time.

Recognizing these challenges, Intuilize offers a structured change management approach and advanced price optimization models designed to empower wholesalers to seamlessly integrate optimized pricing strategies into their Enterprise Resource Planning (ERP) systems.

Our proprietary price optimization model is engineered to determine optimal pricing for every customer and SKU combination. With an average clientele of 5,000 customers and SKU ranges varying between 10,000 to 100,000 items per wholesaler, precision in pricing is imperative.

Impact:

These algorithms are pivotal in enhancing decision-making processes across diverse industries, enabling businesses to formulate data-driven strategies.

The project encompasses a range of data science and ML engineering tasks, offering team members the opportunity to delve into Python-based ML libraries and gain proficiency in classification and prediction models. Initial features utilized in current classification tasks will be provided, with scope for exploring and incorporating additional features to enhance model performance.

Collaboration and Support:

Regular meetings will be conducted with members from Intuilize's Analytics team, providing team members with practical insights into the machine learning lifecycle and its application in real-world scenarios.

This project offers a unique opportunity for team members to contribute to cutting-edge ML applications while honing their skills in data science and model development, with the ultimate goal of driving tangible business outcomes for wholesalers.

Project #1: Price Sensitivity Analyzer

Project Description: This project aims to determine price sensitivity and elasticity across customer segments / product categories to establish optimal price ceilings. By analyzing how different customer groups respond to price changes, we'll develop models that predict purchase behavior at various price points. These insights will enable strategic pricing decisions that optimize gross margin while maintaining customer satisfaction and retention.

Methodology:

- Data Collection and Analysis: Gather historical transaction data including purchase history, pricing variations, and customer responses to previous price changes to identify sensitivity patterns.
- Segmentation: Repurpose existing segmentation & churn predictor models to enhance the model
- Elasticity Modeling: Develop econometric models to calculate price elasticity coefficients for different products and customer segments.
- Price Ceiling Determination: Establish optimal price thresholds for each product-segment combination that maximize profitability without triggering significant demand reduction.
- Testing and Validation: Implement controlled price testing across selected customer segments to validate elasticity models and refine price ceiling recommendations.

Expected Outcomes:

- Optimized Pricing Strategy: Data-driven price points that balance revenue maximization with customer retention.
- Segment-Specific Insights: Detailed understanding of how different customer segments respond to price changes, enabling targeted pricing approaches.
- Competitive Positioning: Strategic price positioning relative to market alternatives based on actual customer sensitivity rather than assumptions.

Model References / Examples:

- <https://docs.datarobot.com/en/docs/api/guide/common-case/python2/elasticity/index.html>
- <https://docs.datarobot.com/en/docs/api/guide/common-case/python2/elasticity/elasticity.html>
- <https://olanipekunayo2012.medium.com/price-elasticity-of-demand-with-python-analysis-c32d70dd5f6>
- <https://www.kaggle.com/code/pratx557/price-elasticity>
- https://github.com/DoubleML/tutorial_materials/blob/main/notebooks/Demand/py_elasticity_analysis.ipynb

Deliverables	Type of work	Activities	Resources	Tech Skills	Priority
Requirements Analysis Document RAD	Requirements analysis	Collaborate closely with the client to comprehensively grasp the requirements, objectives, and expectations - Deliver a comprehensive project scope and requirements document that details the functionality and implementation strategy.	Client Lead and SMEs	System and requirements analysis and documentation	High
Data Model Design and Plan Document DMDP / Data Gathering	Data model planning – what are the fields and structures of the data available for use?	Collaborate with clients and subject matter experts (SMEs) to meticulously identify all available data fields and structures for integration and pre process data. A comprehensive report summarizing the historical sales data collected, including data sources, data quality assessments, and initial findings on purchasing patterns.	Client Lead and SMEs	Data Gathering & Plan Document	Medium
Machine learning predictors – working systems, notebooks, etc. Data Preparation	Data Modeling	<i>Customer Price Sensitivity Segmentation:</i> Apply clustering techniques to group customers by similar purchasing responses to price changes <i>Elasticity Calculation:</i> Determine price elasticity coefficients for each customer/product category and identify cross-price elasticity relationships between related products <i>Price Ceiling Determination:</i> Develop statistical models to identify optimal price thresholds where revenue is maximized without significant demand reduction	Client Lead and SMEs, Project Coach	Data science and machine learning tools	Medium
Machine learning predictors – working systems, notebooks, etc. Modelling	Machine learning modeling and documentation	<i>Price Threshold Pattern Recognition:</i> Apply classification algorithms to identify patterns in customer purchasing data <i>Elasticity Prediction Model</i> Machine learning models (regression, decision trees, etc) <i>Price Ceiling Detection Algorithms:</i> Create models that can automatically detect when customers approach their price ceiling based on historical purchase behavior	Project Coach, Freely available ML tools	Data science and machine learning tools	High
Summary of Findings – presentation and document Presentation	Summarize results of exploration and recommendations	Present results from by testing the proposed customer tiers & predictions - Validate model performance using a holdout test set or cross-validation. - Compare model predictions with actual customer behavior - Conduct model error analysis to identify areas for improvement	Client Lead and SMEs, Project Coach	Presentation	High
Project Handover Documentation		A summary of the project’s goals, objectives, and key deliverables, including the problem statement and the intended impact of the machine learning model. - Technical Documentation: Detailed documentation of the machine learning model including: Deployment Process & Model Monitoring - Project Artifacts: Delivery of all relevant code, scripts, and resources required for the model's operation and future development.		Documentation	High